CFD and Heat Transfer Lab: Experiment 1

External Flow over a NACA 0012 Airfoil in Google Colab

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Colab-first workflow: no local installation, no compiler, no ParaView requirement for basic submission.

What students should learn

Aim

Study how angle of attack changes the velocity field, pressure trend, wake structure, convergence behaviour, and qualitative lift/drag response around a NACA 0012 airfoil.

Strict interpretation rule

The supplied solver is a teaching-level immersed-geometry projection model. The force coefficients are qualitative. They must not be reported as validated aerodynamic data.

Primary learning outcomes

- Identify the role of Reynolds number and angle of attack.
- Connect the pressure-correction algorithm to incompressibility.
- Interpret residual history, wake structure, speed contours, and force trends.

Compulsory comparison

Run $\alpha = -5^\circ, 0^\circ, 5^\circ, 10^\circ$ at $Re = 100$ and compare wake behaviour, pressure-side/suction-side trends, residuals, and qualitative force histories.

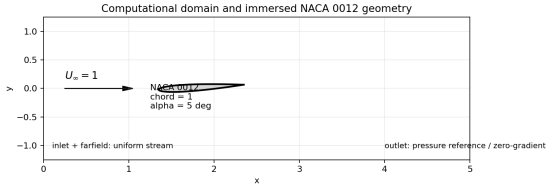
Physical model and geometry

NACA 0012 thickness distribution

$$y_t = 5t(0.2969\sqrt{x} - 0.1260x - 0.3516x^2 + 0.2843x^3 - 0.1036x^4)$$

Laboratory case

Domain	$0 \le x \le 5,\ -1.25 \le y \le 1.25$
Airfoil	NACA 0012, $c = 1$
Default case	$Re = 100$, $\alpha = 5^\circ$
Grid	120×60 in lab mode
Rotation	about $c/4$ point
Velocity scale	$U_\infty = 1$



The airfoil is represented through a smooth immersed mask rather than a body-fitted mesh.

Governing equations used in the teaching solver

Incompressible flow

$$\nabla \cdot \boldsymbol{u} = 0,$$

$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} = -\nabla p + \frac{1}{Re}\nabla^2 \boldsymbol{u} - \frac{\chi}{\eta}\boldsymbol{u}.$$

Meaning of the penalty term

- $\chi = 0$ in fluid and $\chi \approx 1$ inside the airfoil.
- $-\chi \boldsymbol{u}/\eta$ damps velocity inside the solid mask.
- This approximates no-through/no-slip behaviour without a body-fitted grid.

What is deliberately simplified

- No turbulence model.
- No high-order wall treatment.
- No validated lift/drag prediction.
- Smooth immersed mask, not exact body-fitted geometry.

What remains useful

Use it for discretization, projection, residual monitoring, field interpretation, and parameter studies.

Numerical algorithm: projection plus immersed penalization

One pseudo-time iteration

1. Compute upwind convective derivatives.
2. Add central-difference viscous diffusion.
3. Apply immersed penalty inside the airfoil mask.
4. Solve pressure-correction Poisson equation.
5. Correct velocity to reduce divergence.
6. Re-apply boundary conditions and log residuals.

Discrete pressure correction

$$\nabla^2 \phi = \frac{1}{\Delta t} \nabla \cdot \boldsymbol{u}^*,$$

$$\boldsymbol{u}^{n+1} = \boldsymbol{u}^* - \Delta t \nabla \phi, \qquad p^{n+1} = p^n + 0.25\phi.$$

Convergence monitor

$$R_c = \left[\frac{1}{N_f} \sum_{\chi < 0.5} (\nabla \cdot \boldsymbol{u})^2 \right]^{1/2}.$$

Colab workflow and required outputs

Student workflow

1. Open `01_NACA0012_Airfoil.ipynb` in Google Colab.
2. Save a copy in Google Drive.
3. Select `lab` mode or approved `custom` values.
4. Use **Runtime** → **Run all**.
5. Download the generated result ZIP.

Do not submit quick mode

`quick` only checks that the notebook runs; it is not a valid laboratory result.

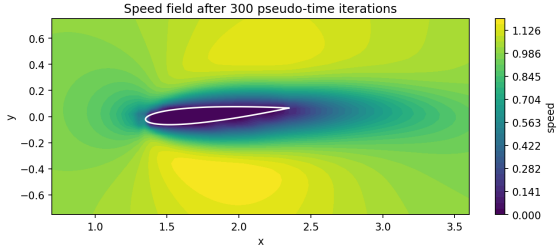
Files produced

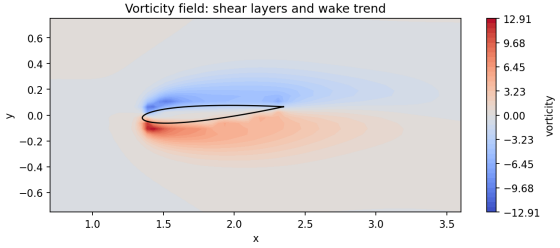
- `history.csv`: residual, speed, qualitative force history.
- `airfoil_final.vtk`: pressure, speed, vorticity, velocity.
- `airfoil_wall.vtk`: airfoil boundary polyline.
- `airfoil_mask.vtk`: immersed solid mask.
- `airfoil_summary.txt`: parameters and final residual.

Minimum report evidence

Residual plot, at least one field contour, tabulated parameter set, and angle-of-attack comparison.

Example output: field interpretation





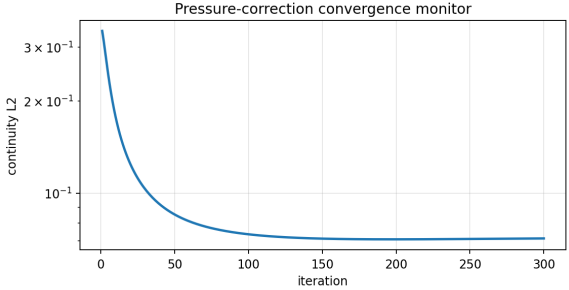
Speed field

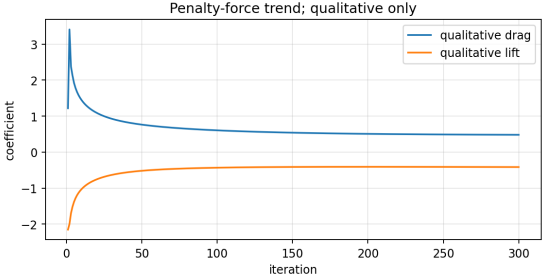
Acceleration around the airfoil and velocity deficit downstream indicate pressure-side/suction-side asymmetry and wake development.

Vorticity field

The strongest gradients appear near the immersed boundary and in the developing wake. Students should compare wake rotation and spread as α changes.

Example output: convergence and qualitative force history





Residual expectation

The continuity residual should decrease from the initial projection error and then approach a pseudo-steady plateau for the selected grid, time step, and Poisson iterations.

Force-coefficient warning

The coefficients are computed from the penalty forcing. They are suitable for trend comparison across α, not for benchmark aerodynamic validation.

Assignment structure and grading focus

Required runs

Case	Re	α	Mode
1	100	-5°	lab/custom
2	100	0°	lab/custom
3	100	5°	lab
4	100	10°	lab/custom

Report must answer

How does α change the wake, pressure/speed asymmetry, residual behaviour, and qualitative lift/drag trend?

Grading emphasis

- Correct parameter reporting.
- Correct plots and labelled axes.
- Reasoned comparison, not screenshot dumping.
- Clear limitation statement.

Common mistakes

Reporting force values as validated C_L, C_D; submitting quick-mode output; omitting residual history; confusing numerical artefact with physics.

Closing technical message for students

What this experiment is really teaching

The purpose is not to produce publishable airfoil aerodynamics. The purpose is to make students see the full CFD chain: geometry representation, boundary conditions, discretization, pressure projection, convergence monitoring, field interpretation, and honest reporting of model limitations.

One-sentence conclusion expected in the report

At fixed $Re = 100$, changing α modifies pressure-side/suction-side asymmetry and wake structure; the supplied solver captures qualitative trends, while its penalty-force coefficients remain teaching-level indicators only.